

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Multiplication of matrices

1. If  $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , find  $A^2$ . (a)  $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  (b)  $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$  (c)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  (d)  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  **ICSE 2025, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

**Answer: (c)**

2. Given  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$  and  $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$ , find (a)  $A + C$  (b)  $B(A + C)$  (c)  $5B$  (d)  $B(A + C) - 5B$ . **ICSE 2025, 4 marks**

SOLUTION

$$(a) A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$$

$$(b) B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$$

$$(c) 5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$$

$$(d) B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$$

$$\text{Answer: (d)} \begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$$

3. If  $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$  and  $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$ , find  $x$ . (a) 2 (b) 4 (c) 8 (d) 10 **ICSE 2024, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4+0 & 4+4 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

**Answer: (c)**

## 2 Solving matrix equations

4.  $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$  and  $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$ . Find  $x$  and  $y$  if  $AB = C$ . **ICSE 2024, 4 marks**

SOLUTION

$$AB = \begin{bmatrix} 4x + 0 & 0 + 0 \\ 4 + y & 0 + 1 \end{bmatrix} = \begin{bmatrix} 4x & 0 \\ 4 + y & 1 \end{bmatrix}$$

Compare with  $C$ :  $4x = 4$ , so  $x = 1$ . Then  $4 + y = x = 1$ , so  $y = -3$ .

**Answer:**  $x = 1, y = -3$

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